

$$V = \mathbb{R}^2 \quad B = \left(\begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 3 \\ 7 \end{pmatrix} \right)$$

$$\vec{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \quad \vec{y} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

$$\vec{x} = a_1 \cdot \begin{pmatrix} 1 \\ 2 \end{pmatrix} + a_2 \cdot \begin{pmatrix} 3 \\ 7 \end{pmatrix} \Rightarrow [\vec{x}]_B = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$$

$$\left(\begin{array}{cc|c} 1 & 3 & x_1 \\ 2 & 7 & x_2 \end{array} \right) \xrightarrow{(-2)} \left(\begin{array}{cc|c} 1 & 3 & x_1 \\ 0 & 1 & x_2 - 2x_1 \end{array} \right) \xrightarrow{(-3)} \left(\begin{array}{cc|c} 1 & 0 & -3x_2 + 7x_1 \\ 0 & 1 & x_2 - 2x_1 \end{array} \right) \Rightarrow *$$

$$[\vec{x}]_B = (-3x_2 + 7x_1, x_2 - 2x_1)^T$$

$$\vec{y} = a_3 \cdot \begin{pmatrix} 1 \\ 2 \end{pmatrix} + a_4 \cdot \begin{pmatrix} 3 \\ 7 \end{pmatrix} \Rightarrow [\vec{y}]_B = \begin{pmatrix} a_3 \\ a_4 \end{pmatrix}$$

$$\left(\begin{array}{cc|c} 1 & 3 & y_1 \\ 2 & 7 & y_2 \end{array} \right) \xrightarrow{(-2)} \left(\begin{array}{cc|c} 1 & 3 & y_1 \\ 0 & 1 & y_2 - 2y_1 \end{array} \right) \xrightarrow{(-3)} \left(\begin{array}{cc|c} 1 & 0 & -3y_2 + 7y_1 \\ 0 & 1 & y_2 - 2y_1 \end{array} \right)$$

$$[\vec{y}]_B = (-3y_2 + 7y_1, y_2 - 2y_1)^T \quad \Downarrow *$$

Podle wzoru 8.50 ze skryptu

$$\langle \vec{x}, \vec{y} \rangle = [\vec{x}]_B^* [\vec{y}]_B$$

$$\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right) = \left[\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \right]_B^* \left[\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right]_B$$

$$= (-3x_2 + 7x_1, x_2 - 2x_1) \begin{pmatrix} -3y_2 + 7y_1 \\ y_2 - 2y_1 \end{pmatrix}$$

$$= (-3x_2 + 7x_1)(-3y_2 + 7y_1) + (x_2 - 2x_1)(y_2 - 2y_1)$$

$$= 9x_2y_2 - 21x_2y_1 - 21x_1y_2 + 49x_1y_1 + x_2y_2 - 2x_2y_1 - 2x_1y_2$$

$$+4x_1y_1 = 10x_2y_2 - 23x_2y_1 - 23x_1y_2 + 53x_1y_1$$